

A Universal Compression Obstruction for Fast $\mathcal{A}$ -Collection Gluing

Candidate negative-answer packet for arXiv:1210.0588

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Status. Candidate full negative answer in the constructional sense of Remark 3.6; likely valid, pending expert review. Every map induced by the normalized-characteristic gluing scheme has Hilbert compression at most $1/3$, so that scheme cannot attain the requested value $1/2$.

Abstract

Baudier asks whether an $\mathcal{A}$ -collection on a connected metric tree can induce an embedding with Hilbert compression $1/2$. We prove a general upper obstruction for the gluing map used immediately before that question. If φ_n are unit-ball-valued, their variation at scale n is at most b_n , and $\sum b_n^p < \infty$, then the unweighted ℓ_p -sum map $\Phi = (\varphi_n - \varphi_n(o))_n$ satisfies

$$\|\Phi(\gamma(t)) - \Phi(\gamma(0))\| = o(t^{1/(p+1)})$$

on every geodesic ray. For an adapted fast $\mathcal{A}$ -collection, Lemma 3.3 gives $b_n^p \leq 2\epsilon_n$ and $\sum \epsilon_n < \infty$. Thus the induced Hilbert map has compression at most $1/3$, giving a negative answer for the paper's construction.

1 Source question and precise scope

The source is F. Baudier, *Quantitative nonlinear embeddings into Lebesgue sequence spaces* [1], arXiv:1210.0588. It defines an adapted fast $\mathcal{A}$ -collection by $r_n = n$ and $\sum_n \epsilon_n < \infty$. Lemma 3.3 then uses normalized characteristic functions and proves the local estimate needed for its gluing map (source PDF p. 25):

- (1) $z \in A(x)$ implies $d(x, z) \leq S$
- (2) $d(x, y) \leq R$ implies $\frac{|A(x) \Delta A(y)|}{|A(x) \cap A(y)|} \leq \varepsilon$

A convenient terminology is needed in order to study property A in its quantitative aspect. Let $\epsilon = (\epsilon_n)_{n \geq 1}$ and $r = (r_n)_{n \geq 1}$. An (ϵ, r) -A collection $\mathcal{A} = (A_n(x))_{(n,x) \in \mathbb{N} \times \mathcal{M}}$ is a collection of compact sets so that for all $n \geq 1$ the collection of sets $(A_n(x))_{x \in \mathcal{M}}$ satisfies the two conditions in the definition of property A with $\varepsilon = \epsilon_n$ and $R = r_n$ for some $S_n > 0$.

A *fast* (ϵ, r) -A collection is an (ϵ, r) -A collection such that $\epsilon = (\epsilon_n)_{n \geq 1}$ is summable. An (ϵ, r) -A collection such that $r_n = n$ is said to be *adapted*.

Since the sets $A_n(x)$ are compact, by condition (1) of property A there exists a number $0 < S_n < \infty$ such that $A_n(x)$ is included in the ball centered at x and of radius S_n . For an A-collection $\mathcal{A} = (A_n(x))_{(n,x) \in \mathbb{N} \times \mathcal{M}}$ denote $\text{rad}_{\mathcal{A}}$ the smallest non-decreasing function S such that for all x and n

$$A_n(x) \subset B_d(x, S(n))$$

and call it the *radial dilation function* of the A-collection. Normalized characteristic functions of fast A-collections produce fundamental maps.

Lemma 3.3. *If a metric measured space $\mathcal{M}$ admits a fast (ϵ, r) -A collection $\mathcal{A}$ then there exist maps*

$\varphi_n: \mathcal{M} \rightarrow S_{\ell_p(\mathcal{M}, \mathbb{R})}$ *satisfying*

- (1) $\|\varphi_n(x) - \varphi_n(y)\|_p \lesssim \epsilon_n^{1/p}$ *whenever* $d(x, y) \leq r_n$
- (2) $\text{supp}(\varphi_n(x)) \subset B(x, \text{rad}_{\mathcal{A}}(n))$

Proof. Let $\mathcal{A} = (A_n(x))_{(n,x) \in \mathbb{N} \times \mathcal{M}}$ be a fast (ϵ, r) -A collection and define

$$\varphi_n(x) = \frac{\chi_{A_n(x)}}{|A_n(x)|^{1/p}}. \text{ Clearly } \varphi_n \in \ell_p(\mathcal{M}, \mathbb{R}), \|\varphi_n(x)\|_p = 1 \text{ and}$$

$$\text{supp}(\varphi_n(x)) = A_n(x) \subset B(x, \text{rad}_{\mathcal{A}}(n))$$

Proposition 3.4 applies this construction to obtain a coarse embedding. For a connected metric tree with an infinite geodesic, Example 3.5 constructs a particular collection, and Remark 3.6 discusses the resulting compression (source PDF pp. 26–27):

$$\begin{aligned}
&\leq \frac{|A_n(x) \Delta A_n(y)| [\max\{|A_n(x)|, |A_n(y)|\} + |A_n(x) \cap A_n(y)|]}{|A_n(x)| |A_n(y)|} \\
&\leq \frac{|A_n(x) \Delta A_n(y)| \cdot 2 \max\{|A_n(x)|, |A_n(y)|\}}{|A_n(x)| |A_n(y)|} \\
&\leq \frac{2|A_n(x) \Delta A_n(y)|}{\min\{|A_n(x)|, |A_n(y)|\}} \\
&\leq \frac{2|A_n(x) \Delta A_n(y)|}{|A_n(x) \cap A_n(y)|} \leq 2\epsilon_n \text{ whenever } d(x, y) \leq r_n
\end{aligned}$$

□

It is by now routine to estimate the compression function in terms of the generalized inverse of the radial dilation function.

Proposition 3.4. *Let $p \geq 1$. If a metric measured space $(\mathcal{M}, d)$ admits an adapted fast A -collection then there is a coarse embedding ϕ from $\mathcal{M}$ into $\ell_p(\mathcal{M}, \mathbb{R})$ such that*

$$\text{rad}_{\mathcal{A}}^-(d(x, y))^{1/p} \lesssim_l \|\phi(x) - \phi(y)\|_p \lesssim_l d(x, y)^{1/p}$$

An typical application of proposition [3.4](#) is presented below.

Example 3.5. *Let $(\mathcal{T}, d)$ be a connected metric tree with an infinite geodesic equipped with the Lebesgue measure. Choose a root $t_0 \in \mathcal{T}$ and denote ω_{t_0} the infinite geodesic ray starting at t_0 . For any point $t \in \mathcal{T}$ there exists a unique geodesic ray ω_t such that $\omega_t \cap \omega_{t_0}$ is infinite. Define $A_n(t)$ to be the geodesic segment of length $r_n \epsilon_n^{-1}$ on the ray ω_t .*

Then $A_n(t) \subset B(t, r_n \epsilon_n^{-1})$ and $d(x, y) \leq r_n$ implies $|A_n(x) \Delta A_n(y)| \leq 2r_n$ while $|A_n(x) \cap A_n(y)| \geq r_n \epsilon_n^{-1}$. Choosing $r_n = n$, $\epsilon_n = \frac{1}{n \log^2(n)}$ one has $\text{rad}_{\mathcal{A}}(n) \leq r_n \epsilon_n^{-1} = n^2 \log(n)^2$ hence $h_{(2,2)} \ll \text{rad}_{\mathcal{A}}^-$. From proposition [3.4](#) it follows that there is a coarse embedding ϕ from $\mathcal{T}$ into $\ell_p(\mathcal{T}, \mathbb{R})$ so that

$$h_{(2,2)}(d_{\mathcal{T}}(x, y))^{1/p} \lesssim_l \|\phi(x) - \phi(y)\|_p \lesssim_l d_{\mathcal{T}}(x, y)^{1/p}.$$

Remark 3.6. When t is large, $t^s \leq h_{(2,2)}(t) \leq t$ for every $0 < s < 1/2$ and at the embedding provide a lower bound $\frac{1}{4}$ for the Hilbert compression. The best value that can be achieved from our embedding is $\frac{1}{2}$ but the Hilbert compression of a tree is 1 (see for instance [9](#)). It is not surprising that our embedding cannot achieve the best compression since our approach can be applied in a wide setting and requires mild geometric properties for

the domain space. Nevertheless it would be interesting to know whether or not it is possible to construct an A-collection inducing an embedding with compression $\frac{1}{2}$.

3.2. Metric locally compact amenable groups.

The exact question is whether it is possible “to construct an A-collection inducing an embedding with compression $1/2$.” In context, “inducing” refers to the normalized-characteristic, unweighted ℓ_p -sum construction of Lemma 3.3 and Proposition 3.4: this is the map the remark calls “our embedding.” We answer that constructional question negatively. We do not assert an upper bound on arbitrary embeddings of the tree, whose Hilbert compression is indeed 1.

2 Proof intuition

Fastness forces the local variations of the coordinate maps to be p -summable. Consider two points distance t apart on a geodesic ray and split the glued coordinates at m . In the first m coordinates, the difference of two unit vectors costs at most 2 per coordinate. In every later coordinate $n > m$, subdivide the ray segment into at most $\lceil t/n \rceil$ pieces of length at most n ; summing the local variation along that chain controls the coordinate difference. The head therefore costs $O(m)$ in the p -th power, while the tail costs at most $O((t/m)^p)$ times a summable tail. Balancing at $m \asymp t^{p/(p+1)}$ gives exponent $1/(p+1)$. Since the summable tail actually tends to zero, a two-limit argument improves the estimate to little- o .

3 The gluing obstruction

For Banach spaces $(X_n)_{n \geq 1}$, write $(\bigoplus_n X_n)_p$ for their ℓ_p -sum. The compression exponent of a particular map is the supremum of exponents α for which its norm difference is bounded below by a positive multiple of $d(x, y)^\alpha$ at all sufficiently large distances.

Theorem 1 (Ray-chain obstruction). *Let $1 \leq p < \infty$, and let a metric space (M, d) contain an isometric geodesic ray $\gamma : [0, \infty) \rightarrow M$. Suppose that $\varphi_n : M \rightarrow X_n$ take values in the closed unit balls and that numbers $b_n \geq 0$ satisfy*

$$\|\varphi_n(x) - \varphi_n(y)\| \leq b_n \quad \text{whenever } d(x, y) \leq n, \quad \sum_{n=1}^{\infty} b_n^p < \infty. \quad (1)$$

For a basepoint $o \in M$, define

$$\Phi(x) = (\varphi_n(x) - \varphi_n(o))_{n \geq 1}.$$

Then Φ is well defined and, uniformly in $s \geq 0$,

$$\|\Phi(\gamma(s+t)) - \Phi(\gamma(s))\| = o(t^{1/(p+1)}) \quad (t \rightarrow \infty). \quad (2)$$

In particular, the compression exponent of Φ is at most $1/(p+1)$.

Proof. For fixed $x \in M$, condition (1) applies to the pair (x, o) for every integer $n \geq d(x, o)$. The tail of $(\varphi_n(x) - \varphi_n(o))_n$ is therefore dominated in p -th power by $\sum b_n^p$, and its head is finite. Thus $\Phi(x)$ belongs to the ℓ_p -sum.

Fix $s \geq 0$, $t \geq 1$, and put $x = \gamma(s)$, $y = \gamma(s+t)$. Let $m \geq 1$ be an integer and set $E_m = \sum_{n>m} b_n^p$. Since the coordinate maps take values in unit balls,

$$\sum_{n \leq m} \|\varphi_n(x) - \varphi_n(y)\|^p \leq 2^p m. \quad (3)$$

For $n > m$, partition the ray segment from x to y into $L = \lceil t/n \rceil$ consecutive subsegments, each of length at most n . Applying (1) to every adjacent pair and using the triangle inequality gives

$$\|\varphi_n(x) - \varphi_n(y)\| \leq \left\lceil \frac{t}{n} \right\rceil b_n \leq \left(1 + \frac{t}{m}\right) b_n.$$

The basepoint terms cancel in $\Phi(x) - \Phi(y)$, so (3) and the last estimate imply

$$\begin{aligned} \|\Phi(x) - \Phi(y)\|^p &\leq 2^p m + \left(1 + \frac{t}{m}\right)^p E_m \\ &\leq 2^p m + 2^{p-1} \left(1 + \left(\frac{t}{m}\right)^p\right) E_m. \end{aligned} \quad (4)$$

Taking $m = \lceil t^{p/(p+1)} \rceil$ already gives the big- O version of (2).

For the sharper statement, fix $0 < \delta < 1$ and take $m = \lceil \delta t^{p/(p+1)} \rceil$. Divide (4) by $t^{p/(p+1)}$. As $t \rightarrow \infty$, one has $m \rightarrow \infty$ and hence $E_m \rightarrow 0$. The tail terms tend to zero, while the limsup of the head term is at most $2^p \delta$. Thus

$$\limsup_{t \rightarrow \infty} \frac{\|\Phi(\gamma(s+t)) - \Phi(\gamma(s))\|^p}{t^{p/(p+1)}} \leq 2^p \delta.$$

The estimate is independent of s . Letting $\delta \downarrow 0$ proves (2). Since the ray contains pairs at every arbitrarily large distance, no coarse lower power with exponent greater than $1/(p+1)$ can hold for Φ . $\square$

4 Negative answer for the source construction

Corollary 2. *Let $\mathcal{T}$ be a connected metric tree with an infinite geodesic. No adapted fast $\mathcal{A}$ -collection can, through the normalized-characteristic gluing map of Lemma 3.3 and Proposition 3.4 of the source, induce a Hilbert-space embedding with compression $1/2$. Every such induced map has compression exponent at most $1/3$.*

Proof. An infinite geodesic contains a geodesic ray. For an adapted fast $\mathcal{A}$ -collection, the source defines

$$\varphi_n(x) = \frac{\chi_{A_n}(x)}{|A_n(x)|^{1/p}}, \quad \|\varphi_n(x)\|_p = 1,$$

and proves

$$\|\varphi_n(x) - \varphi_n(y)\|_p^p \leq 2\epsilon_n \quad \text{whenever } d(x, y) \leq r_n = n.$$

Set $b_n = (2\epsilon_n)^{1/p}$. Fastness says $\sum_n \epsilon_n < \infty$, so $\sum_n b_n^p = 2 \sum_n \epsilon_n < \infty$. Theorem 1 applies. At $p = 2$ it gives compression exponent at most $1/(2+1) = 1/3$, which rules out $1/2$. $\square$

The obstruction is independent of radial dilation and of the particular ray-segment collection in Example 3.5. It therefore shows that optimizing the support radii alone cannot reach the value suggested in Remark 3.6: the summable local-variation requirement and the unweighted gluing operation already impose a stricter ceiling.

5 Verification, limitations, and novelty

The proof uses only three ingredients: the unit-ball bound, a chain inside an actual geodesic ray, and the vanishing tails of a summable sequence. The basepoint cancels from differences. Compact supports and radial dilation, which establish the source’s coarse lower bound, are irrelevant to the new upper bound.

The scope qualification is important. The theorem concerns exactly the map $\Phi = (\varphi_n - \varphi_n(o))_n$ appearing in the source. It does not rule out a materially different weighted or nonlinear map which merely uses an $\mathcal{A}$ -collection as auxiliary information, and it does not bound the tree’s intrinsic Hilbert compression.

Bounded searches on 11 August 2026 covered the run’s four cheap indexes; the exact arXiv id, title, and question wording; title/citation variants; and combinations of adapted fast $\mathcal{A}$ -collections, radial dilation, normalized characteristic maps, trees, compression 1/2, and compression 1/3. No later paper explicitly answering Remark 3.6 and no matching universal gluing obstruction were located. Novelty is provisional.

Human review should first confirm the contextual reading of “inducing.” The mathematical audit should then check the ray-chain estimate and the two-limit little- o optimization. If a broader meaning of “inducing” was intended, the theorem should be classified as a sharp obstruction for the paper’s explicit method rather than as a claim about all conceivable $\mathcal{A}$ -collection-based constructions.

References

- [1] F. Baudier, *Quantitative nonlinear embeddings into Lebesgue sequence spaces*, arXiv:1210.0588.